UC SAN DIEGO / ECE 251C

13
Vanishing moments and wavelet filter design

Coefficient decay, spectral zeros, and root allocation

How do zeros of a filter control the size of wavelet coefficients?

13.1 From exact cancellation to small coefficients

In Lecture 12, a wavelet coefficient vanished whenever its analyzing wavelet lay entirely inside a polynomial piece of sufficiently low degree. A real signal need not be exactly polynomial. The useful question is now quantitative: how large is the coefficient when the signal is only *approximately* polynomial on that support?

Recall the design tradeoff in the real, compactly supported, orthonormal dyadic MRA setting. A wavelet with p vanishing moments has support span at least $2p - 1$, and its low-pass filter has length at least $2p$. The Daubechies family attains these bounds. Filter length here is the span from the first to the last tap, including any intervening zeros. These orthonormal bounds do not apply to arbitrary biorthogonal pairs.

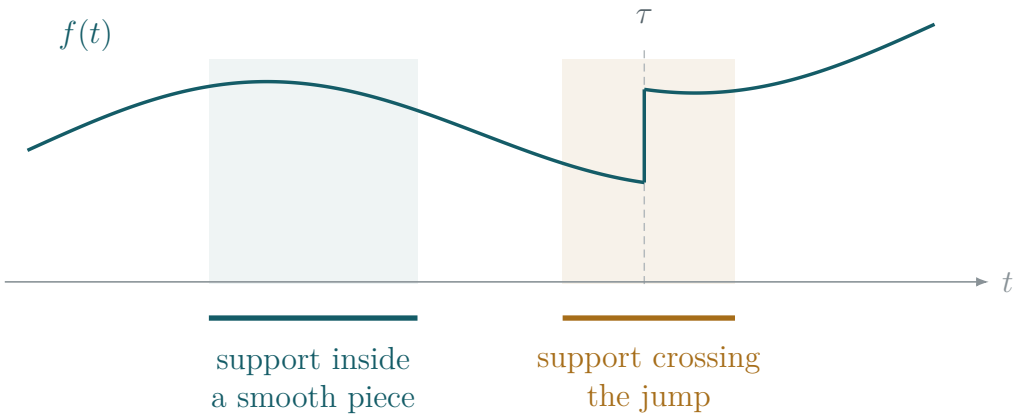


Figure 13.1: The estimate we will derive is local. It applies when the entire wavelet support, together with the Taylor expansion point, lies inside an interval of sufficient smoothness. Crossing a jump changes the problem.

13.2 The local polynomial model

Let $p \geq 1$ be an integer and let $\psi \in L^2(\mathbb{R})$ be compactly supported, with $\|\psi\|_{L^2} = 1$. Suppose that

$$\int_{\mathbb{R}} u^m \psi(u)\, \mathrm{d}u = 0, \qquad m = 0, \ldots, p-1. \tag{13.1}$$

As before, "p vanishing moments" means *at least* p. For a signal $f \in L^2(\mathbb{R})$, write

$$\psi_{i,n}(t) = 2^{i/2}\psi(2^i t - n), \qquad d_i[n] = \int_{\mathbb{R}} f(t)\psi_{i,n}(t)^*\, \mathrm{d}t.$$

The second factor is conjugated. For fixed $i, n \in \mathbb{Z}$, set $t_0 = 2^{-i}n$. Assume that f has p continuous derivatives on an interval $\mathcal{I}$ containing t_0 and $\operatorname{supp}\psi_{i,n}$, and that $|f^{(p)}(t)| \leq M$ there.

The Taylor polynomial of degree at most $p-1$ is

$$T_{p-1}(t) = \sum_{m=0}^{p-1} \frac{f^{(m)}(t_0)}{m!}(t - t_0)^m. \tag{13.2}$$

For $p = 1$, it retains the local level; for $p = 2$, the level and slope; for $p = 3$, it also retains curvature. Define the remainder by

$$R_p(t) = f(t) - T_{p-1}(t).$$

We do not assume that f has a convergent infinite Taylor series. A finite polynomial and a controlled remainder are enough.

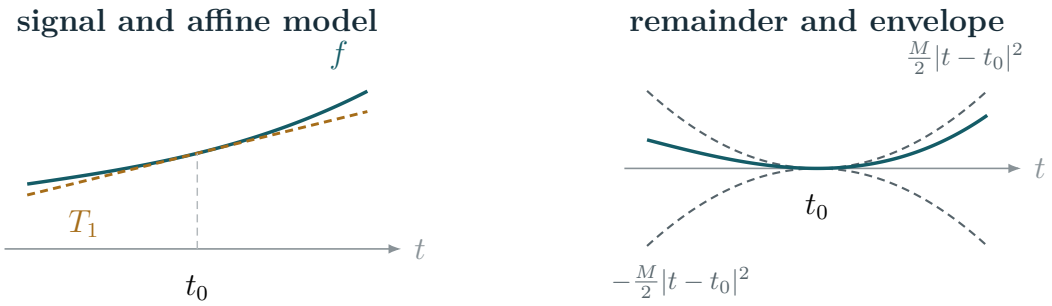


Figure 13.2: A local example with $v = t - t_0$: $f = 1 + 0.8v + v^2 + v^3$, $T_1 = 1 + 0.8v$, and $|v| \leq 0.3$. Since $|f''| \leq 3.8$, the signed remainder lies between $\pm(3.8/2)v^2$. The two panels use different vertical scales.

13.3 Bounding the Taylor remainder

Taylor's integral formula is

$$R_p(t) = \frac{1}{(p-1)!}\int_{t_0}^{t}(t-s)^{p-1}f^{(p)}(s)\,\mathrm{d}s. \tag{13.3}$$

Here is a derivation. For $p = 1$, the formula is the fundamental theorem of calculus. For $p \geq 2$, denote the right-hand side by $Q_p(t)$. Integration by parts gives

$$\begin{aligned} Q_p(t) &= \left[\frac{(t-s)^{p-1}}{(p-1)!}f^{(p-1)}(s)\right]_{t_0}^{t} + \frac{1}{(p-2)!}\int_{t_0}^{t}(t-s)^{p-2}f^{(p-1)}(s)\,\mathrm{d}s \\ &= Q_{p-1}(t) - \frac{f^{(p-1)}(t_0)}{(p-1)!}(t-t_0)^{p-1}. \end{aligned}$$

Starting from $Q_1(t) = f(t) - f(t_0)$, each step subtracts the next Taylor term. This proves (13.3).

If $t \geq t_0$, then

$$\begin{aligned} |R_p(t)| &\leq \frac{1}{(p-1)!}\int_{t_0}^{t}(t-s)^{p-1}|f^{(p)}(s)|\,\mathrm{d}s \\ &\leq \frac{M}{(p-1)!}\int_{t_0}^{t}(t-s)^{p-1}\,\mathrm{d}s = \frac{M}{p!}(t-t_0)^p. \end{aligned}$$

For $t < t_0$, reverse the integration limits before taking absolute values:

$$|R_p(t)| \leq \frac{M}{(p-1)!}\int_{t}^{t_0}(s-t)^{p-1}\,\mathrm{d}s = \frac{M}{p!}(t_0-t)^p.$$

Thus both directions give the same bound:

$$\boxed{|f(t) - T_{p-1}(t)| \leq \frac{M}{p!}|t-t_0|^p, \qquad t \in \mathcal{I}.} \tag{13.4}$$

The factor $1/p$ comes from integrating a power of degree $p-1$; together with $(p-1)!$, it produces $p!$.

13.4 The coefficient sees only the remainder

We first verify that vanishing moments survive translation and dilation. For $0 \le m < p$, use $u = 2^i t - n$, so that

$$t - t_0 = 2^{-i}u, \qquad \mathrm{d}t = 2^{-i}\,\mathrm{d}u.$$

Then

$$\begin{aligned}\int_{\mathbb{R}} (t-t_0)^m \psi_{i,n}(t)^*\,\mathrm{d}t &= \int_{\mathbb{R}} (2^{-i}u)^m\, 2^{i/2}\psi(u)^*\, 2^{-i}\,\mathrm{d}u \\ &= 2^{-i(m+1/2)} \left(\int_{\mathbb{R}} u^m \psi(u)\,\mathrm{d}u\right)^* = 0.\end{aligned}$$

Every term of (13.2) is therefore canceled:

$$\int_{\mathbb{R}} T_{p-1}(t)\psi_{i,n}(t)^*\,\mathrm{d}t = 0.$$

These integrals involve only the compact wavelet support. We are not asserting that a polynomial on the whole line belongs to $L^2(\mathbb{R})$.

Now add and subtract the Taylor polynomial inside the coefficient:

$$\begin{aligned}d_i[n] &= \int_{\mathbb{R}} \big[f(t) - T_{p-1}(t) + T_{p-1}(t)\big]\psi_{i,n}(t)^*\,\mathrm{d}t \\ &= \int_{\operatorname{supp}\psi_{i,n}} \big[f(t) - T_{p-1}(t)\big]\psi_{i,n}(t)^*\,\mathrm{d}t.\end{aligned}$$

The triangle inequality and (13.4) imply

$$\begin{aligned}|d_i[n]| &\le \int_{\operatorname{supp}\psi_{i,n}} |f(t) - T_{p-1}(t)|\,|\psi_{i,n}(t)|\,\mathrm{d}t \\ &\le \frac{M}{p!}\int_{\mathbb{R}} |t-t_0|^p\, 2^{i/2}|\psi(2^i t - n)|\,\mathrm{d}t. \qquad (13.5)\end{aligned}$$

Extending the last integral to $\mathbb{R}$ changes nothing: the wavelet vanishes outside its support. Smoothness is needed only on $\mathcal{I}$.

> **Cancellation is automatic**
> The DWT does not fit and subtract a Taylor polynomial. The wavelet's moments make the polynomial's contribution zero without ever computing it.

13.5 Every factor in the scale-dependent estimate

In (13.5), make the increasing affine substitution

$$u = 2^i t - n, \qquad t = 2^{-i}(u+n), \qquad t_0 = 2^{-i} n.$$

Consequently,

$$|t - t_0|^p = 2^{-ip}|u|^p, \qquad \psi(2^i t - n) = \psi(u), \qquad \mathrm{d}t = 2^{-i}\,\mathrm{d}u.$$

The integration limits stay $-\infty$ and $+\infty$. Substitution gives

$$\begin{aligned} |d_i[n]| &\le \frac{M}{p!} \int_{\mathbb{R}} 2^{-ip}|u|^p\, 2^{i/2}|\psi(u)|\, 2^{-i}\,\mathrm{d}u \\ &= \frac{M}{p!}\, 2^{-ip+i/2-i} \int_{\mathbb{R}} |u|^p|\psi(u)|\,\mathrm{d}u \\ &= \frac{M}{p!}\, 2^{-i(p+1/2)} \int_{\mathbb{R}} |u|^p|\psi(u)|\,\mathrm{d}u. \end{aligned}$$

The last equality is an exact change of variables in the *upper bound*, not a claim that the coefficient equals this bound.

Theorem 13.1 (Local decay of wavelet coefficients)**.** Under the assumptions of Section 13.2,

$$\boxed{|d_i[n]| \le C_{\psi,p} M\, 2^{-i(p+1/2)}, \qquad C_{\psi,p} = \frac{1}{p!} \int_{\mathbb{R}} |u|^p|\psi(u)|\,\mathrm{d}u.} \tag{13.6}$$

The preceding calculation proves the theorem. Its constant is finite: on the compact support, $|u|^p$ is bounded, and Cauchy–Schwarz gives $\|\psi\|_{L^1} \le |\operatorname{supp}\psi|^{1/2}\|\psi\|_{L^2}$. The constant depends on ψ and p, but not on the scale or location.

$$\underbrace{2^{-ip}}_{\substack{\text{Taylor}\\ \text{error}}} \qquad \underbrace{2^{i/2}}_{\substack{\text{unit-energy}\\ \text{amplitude}}} \qquad \underbrace{2^{-i}}_{\substack{\text{integration}\\ \text{variable}}} = 2^{-i(p+1/2)}.$$

Increasing i shortens the analyzing interval. The order-p polynomial approximation improves by 2^{-p} per refinement; the remaining $2^{-1/2}$ comes from the coefficient normalization.

13.6 Rapid decay is not the same as an exact zero

For a uniform derivative bound M over the supports under consideration, the envelope in (13.6) decreases geometrically in i. This is exponential decay in the *scale index*, or algebraic decay in the inverse width 2^i.

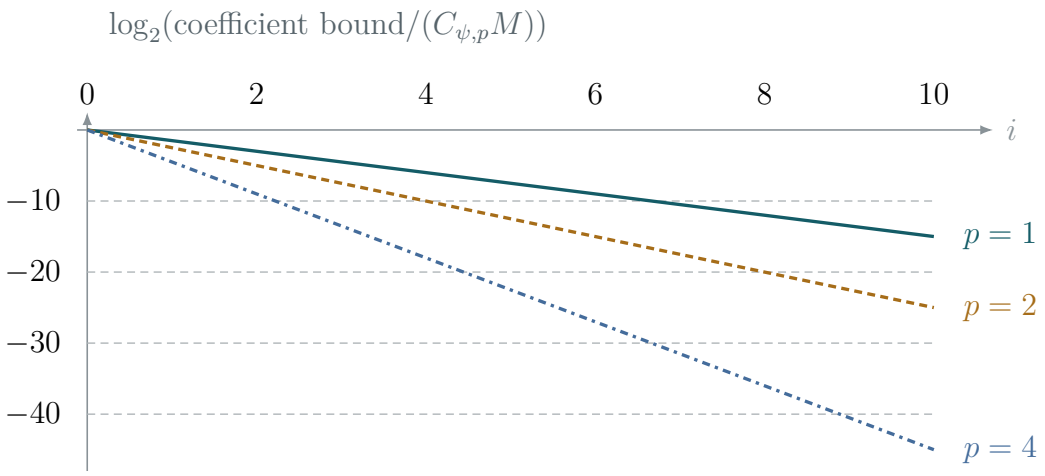


Figure 13.3: Normalized upper bounds, not measured coefficients. Their base-two logarithms have slope $-(p + 1/2)$. Different wavelets and derivative orders generally have different constants $C_{\psi,p}M$.

For example, the scale factor alone at $i = 10$ is

p	1	2	4
$2^{-10(p+1/2)}$	2^{-15}	2^{-25}	2^{-45}
	3.05×10^{-5}	2.98×10^{-8}	2.84×10^{-14}

These numbers must still be multiplied by $C_{\psi,p}M$. There is no universal scale beyond which all smooth-region coefficients are numerically zero: rescaling the signal rescales M and every coefficient.

For a tolerance $\varepsilon > 0$, a particular coefficient is guaranteed to satisfy $|d_i[n]| \le \varepsilon$ whenever

$$C_{\psi,p}M\, 2^{-i(p+1/2)} \le \varepsilon.$$

This connects the bound to deciding which details are negligible. It does not assert monotonicity of individual coefficients between scales, or supply the bound on supports crossing a singularity. A wavelet with p moments can use this argument with any smaller positive order for which the signal has the required derivative bound.

13.7 A worked example: Haar on a ramp

For Haar,

$$\psi(u) = \mathbf{1}_{[0,1/2)}(u) - \mathbf{1}_{[1/2,1)}(u), \qquad \int \psi(u)\,\mathrm{d}u = 0.$$

Its first moment is not zero:

$$\int_0^1 u\psi(u)\,\mathrm{d}u = \int_0^{1/2} u\,\mathrm{d}u - \int_{1/2}^1 u\,\mathrm{d}u = \frac{1}{8} - \frac{3}{8} = -\frac{1}{4}.$$

Suppose $f(t) = A + Bt$ on the wavelet support. The same substitution gives

$$\begin{aligned} d_i[n] &= 2^{-i/2}\int_0^1 \big[A + B2^{-i}(u+n)\big]\psi(u)\,\mathrm{d}u \\ &= 2^{-i/2}(A + B2^{-i}n)\underbrace{\int_0^1 \psi(u)\,\mathrm{d}u}_{0} + B2^{-3i/2}\underbrace{\int_0^1 u\psi(u)\,\mathrm{d}u}_{-1/4} \\ &= -\frac{B}{4}\,2^{-3i/2}. \end{aligned}$$

Thus Haar removes the level but responds to the slope. In particular, $f(t) = t$ gives $d_i[n] = -2^{-3i/2}/4$.

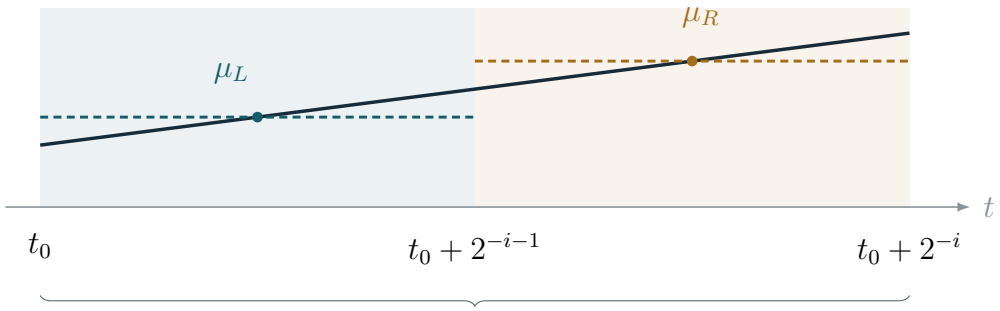


Figure 13.4: Haar compares the average over the left half of its support with the average over the right half. A positive slope makes the first average smaller, explaining the negative coefficient in our sign convention.

If these averages are μ_L, μ_R, each half-integral equals 2^{-i-1} times its average. Hence $d_i[n] = 2^{-i/2-1}(\mu_L - \mu_R)$. For the affine signal, $\mu_L - \mu_R = -B2^{-i-1}$, which gives the same result. A wavelet with two vanishing moments cancels the affine piece exactly.

13.8 The two-scale equations in the Fourier domain

The estimate explains why moments are useful. We next express them as conditions on the filters, so they can be designed before the wavelet is known explicitly.

Assume now that ϕ, ψ are compactly supported functions from an orthonormal MRA, with $\int \phi(t)\,\mathrm{d}t = 1$, and FIR synthesis filters g_0, g_1. The two-scale equations are

$$\phi(t) = \sqrt{2}\sum_n g_0[n]\phi(2t-n), \qquad \psi(t) = \sqrt{2}\sum_n g_1[n]\phi(2t-n). \tag{13.7}$$

Recall our Fourier conventions:

$$\widehat{f}(\omega) = \int_{\mathbb{R}} f(t)\mathrm{e}^{-\mathrm{j}\omega t}\,\mathrm{d}t, \qquad \widehat{g}_\ell(\omega) = \sum_n g_\ell[n]\mathrm{e}^{-\mathrm{j}\omega n} = G_\ell(\mathrm{e}^{\mathrm{j}\omega}).$$

For each term in (13.7), $u = 2t - n$ gives

$$\begin{aligned}\int_{\mathbb{R}} \sqrt{2}\,\phi(2t-n)\mathrm{e}^{-\mathrm{j}\omega t}\,\mathrm{d}t &= \frac{1}{\sqrt{2}}\int_{\mathbb{R}} \phi(u)\mathrm{e}^{-\mathrm{j}(\omega/2)(u+n)}\,\mathrm{d}u \\ &= \frac{1}{\sqrt{2}}\mathrm{e}^{-\mathrm{j}\omega n/2}\widehat{\phi}(\omega/2).\end{aligned}$$

Summing and replacing ω by 2ω yields

$$\boxed{\widehat{\phi}(2\omega) = \frac{1}{\sqrt{2}}\widehat{g}_0(\omega)\widehat{\phi}(\omega), \qquad \widehat{\psi}(2\omega) = \frac{1}{\sqrt{2}}\widehat{g}_1(\omega)\widehat{\phi}(\omega).} \tag{13.8}$$

With $g_1[n] = (-1)^n g_0^*[K-n]$ and K odd, the conjugate-mirror identity is

$$\widehat{g}_1(\omega) = -\mathrm{e}^{-\mathrm{j}K\omega}\widehat{g}_0(\omega+\pi)^*.$$

Therefore

$$\widehat{\psi}(2\omega) = -\frac{\mathrm{e}^{-\mathrm{j}K\omega}}{\sqrt{2}}\widehat{g}_0(\omega+\pi)^*\widehat{\phi}(\omega). \tag{13.9}$$

The factor $\widehat{\phi}(\omega)$ remains present. It is nonzero near the origin because it is continuous and $\widehat{\phi}(0) = 1$.

13.9 Vanishing moments are spectral zeros

Compact support permits differentiation under the Fourier integral: every polynomial multiple of ψ is integrable. Thus

$$\widehat{\psi}'(\omega) = -\mathrm{j}\int_{\mathbb{R}} t\psi(t)\mathrm{e}^{-\mathrm{j}\omega t}\,\mathrm{d}t,$$
$$\widehat{\psi}^{(m)}(\omega) = (-\mathrm{j})^m\int_{\mathbb{R}} t^m\psi(t)\mathrm{e}^{-\mathrm{j}\omega t}\,\mathrm{d}t.$$

At zero frequency this becomes

$$\widehat{\psi}^{(m)}(0) = (-\mathrm{j})^m\int_{\mathbb{R}} t^m\psi(t)\,\mathrm{d}t. \tag{13.10}$$

Since $(-\mathrm{j})^m \neq 0$, the first p moments vanish exactly when the value and first $p-1$ derivatives of $\widehat{\psi}$ vanish at zero.

A nonzero factor does not change this order. In (13.9), set $F = BC$, where

$$F(\omega) = \widehat{\psi}(2\omega), \quad B(\omega) = -\frac{\mathrm{e}^{-\mathrm{j}K\omega}}{\sqrt{2}}\widehat{\phi}(\omega), \quad C(\omega) = \widehat{g}_0(\omega+\pi)^*.$$

Here $B(0) = -1/\sqrt{2} \neq 0$. The product rule gives

$$F^{(m)}(0) = \sum_{r=0}^{m}\binom{m}{r}B^{(r)}(0)C^{(m-r)}(0).$$

If $C^{(r)}(0) = 0$ for $r < p$, then $F^{(m)}(0) = 0$ for $m < p$. Conversely, $F(0) = 0$ forces $C(0) = 0$. Inductively, after the lower derivatives of C vanish, the formula reduces to $F^{(m)}(0) = B(0)C^{(m)}(0)$, forcing the next derivative to vanish as well.

Finally, $F^{(m)}(0) = 2^m\widehat{\psi}^{(m)}(0)$ and $C^{(m)}(0) = \widehat{g}_0^{(m)}(\pi)^*$. We have proved

$$\boxed{\begin{array}{c}\psi \text{ has } p \text{ vanishing moments}\\ \iff \quad \widehat{g}_1^{(m)}(0) = 0 \quad (0 \le m < p)\\ \iff \quad \widehat{g}_0^{(m)}(\pi) = 0 \quad (0 \le m < p).\end{array}} \tag{13.11}$$

The high-pass version follows from the second identity in (13.8) by the same nonzero-factor argument. This proof allows complex filters.

13.10 From frequency derivatives to factors of the filter

Differentiating the finite DTFT sum gives

$$\widehat{g}_0^{(m)}(\pi) = (-\mathrm{j})^m \sum_n (-1)^n n^m g_0[n].$$

Hence the moment conditions can be checked directly on the low-pass taps:

$$\boxed{\sum_n (-1)^n n^m g_0[n] = 0, \qquad m = 0, \dots, p-1.} \tag{13.12}$$

For $m = 0$, the convention is $n^0 = 1$, including at zero. This first condition is the familiar alternating tap sum. Higher moments give higher-order cancellation at the same frequency.

For a filter supported in $\{0, \dots, K\}$, set $P(v) = \sum_{n=0}^{K} g_0[n] v^n$, so $G_0(z) = P(z^{-1})$. If $P(v) = (v+1)^r Q(v)$ with $Q(-1) \neq 0$, then

$$P(\mathrm{e}^{-\mathrm{j}(\pi+\eta)}) = (1 - \mathrm{e}^{-\mathrm{j}\eta})^r Q(-\mathrm{e}^{-\mathrm{j}\eta}).$$

Since $1 - \mathrm{e}^{-\mathrm{j}\eta} = \mathrm{j}\eta + O(\eta^2)$, this has a zero of order exactly r at $\eta = 0$. Thus (13.11) is equivalent to

$$\boxed{G_0(z) = (1 + z^{-1})^p Q_0(z),} \tag{13.13}$$

where Q_0 is a polynomial in z^{-1}. Equivalently, G_0 has a zero at $z = -1$ of multiplicity at least p. The high-pass filter has the corresponding factor $(1 - z^{-1})^p$ at $z = 1$.

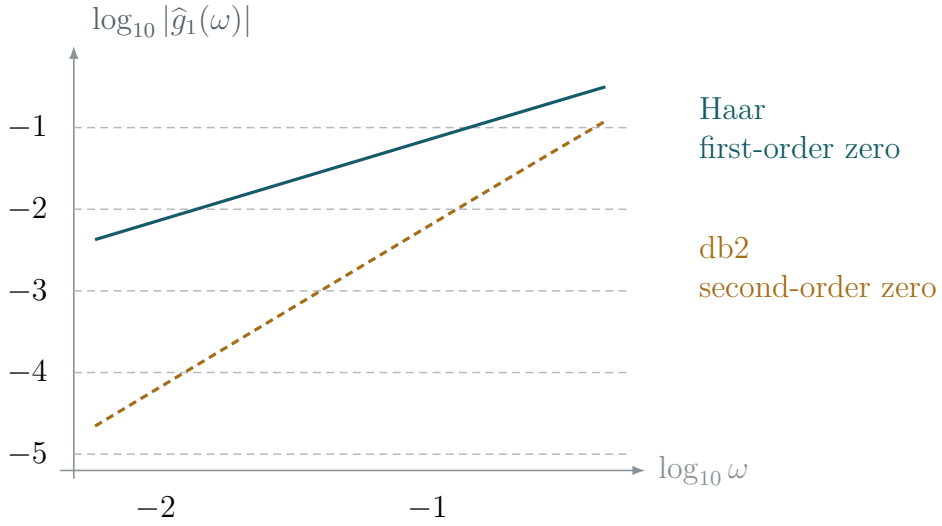


Figure 13.5: Low-frequency high-pass magnitudes for Haar and db2. A logarithmic plot reveals first- and second-order zeros: the responses behave as constant multiples of $|\omega|$ and $|\omega|^2$ near zero.

13.11 Product filters and the causal indexing convention

The remaining examples use the causal product-filter convention from Lecture 9. We make this change explicit. Previously, the analysis filter associated with $g_0[0], \ldots, g_0[K]$ had transform

$$H_0^{\text{zero}}(z) = [G_0(1/z^*)]^* .$$

Delaying it by K makes it causal. In the examples below, the symbol H_0 denotes this delayed filter:

$$H_0(z) = z^{-K} [G_0(1/z^*)]^* . \tag{13.14}$$

For real taps this is $H_0(z) = z^{-K} G_0(z^{-1})$, or $h_0[n] = g_0[K-n]$. This delay changes phase and indexing, but not magnitudes or vanishing-moment orders.

Define $P_0(z) = H_0(z)G_0(z)$. In the orthogonal case,

$$P_0(\mathrm{e}^{\mathrm{j}\omega}) = \mathrm{e}^{-\mathrm{j}K\omega} |G_0(\mathrm{e}^{\mathrm{j}\omega})|^2 .$$

Power complementarity and odd K give

$$\boxed{P_0(z) - P_0(-z) = 2z^{-K}.} \tag{13.15}$$

For FIR filters, equality on the unit circle is also equality of the Laurent polynomials: multiply by a sufficiently large power of z, then use that a nonzero polynomial cannot have infinitely many roots.

More generally, any FIR factorization $P_0 = H_0 G_0$ satisfying (13.15) gives a perfect-reconstruction bank by setting

$$H_1(z) = -G_0(-z), \qquad G_1(z) = H_0(-z).$$

Indeed,

$$\begin{aligned} G_0(z)H_0(-z) + G_1(z)H_1(-z) &= 0, \\ G_0(z)H_0(z) + G_1(z)H_1(z) &= P_0(z) - P_0(-z) = 2z^{-K}. \end{aligned}$$

The input–output relation is therefore $Y(z) = z^{-K} X(z)$. Both high-pass signs may be reversed together without changing reconstruction. Factorization alone guarantees this reconstruction identity, not orthogonality.

13.12 Haar: divide two zeros into one and one

Start with the three-tap product filter

$$P_0(z) = \frac{1}{2}(1 + z^{-1})^2 = \frac{1}{2} + z^{-1} + \frac{1}{2}z^{-2}. \tag{13.16}$$

Its odd coefficient is the unit tap at index one, so

$$P_0(z) - P_0(-z) = 2z^{-1}.$$

A balanced factorization is

$$G_0(z) = H_0(z) = \frac{1}{\sqrt{2}}(1 + z^{-1}).$$

Each factor has DC gain $\sqrt{2}$ and one zero at $z = -1$. The construction on the preceding page gives

$$G_1(z) = \frac{1}{\sqrt{2}}(1 - z^{-1}), \qquad H_1(z) = -\frac{1}{\sqrt{2}}(1 - z^{-1}).$$

Thus the complete causal bank reconstructs with a one-sample delay. This is the same Haar transform as before, with the analysis filters delayed.

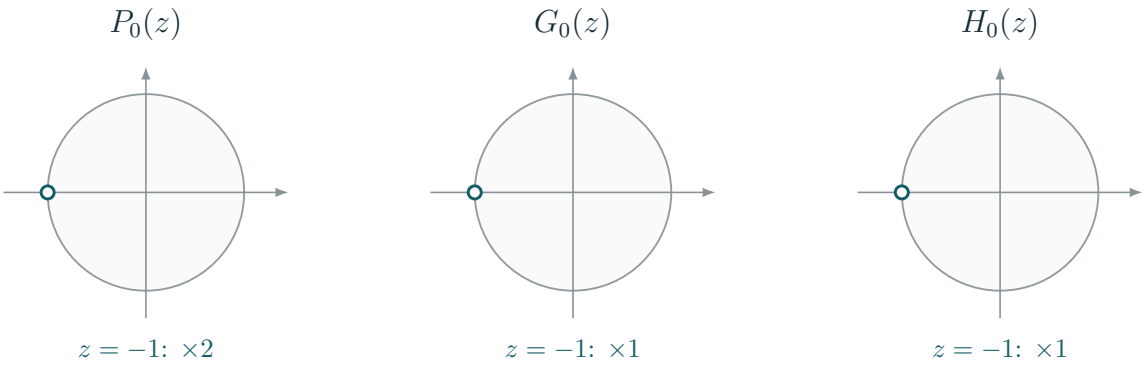


Figure 13.6: Factorization of the Haar product: $P_0 = G_0 H_0$. Open circles mark zeros, and $\times r$ below a panel gives the multiplicity at $z = -1$. The gray circle is $|z| = 1$; the horizontal and vertical axes are $\mathrm{Re}\, z$ and $\mathrm{Im}\, z$, with equal scales. Poles at the origin are omitted throughout these zero diagrams.

The product has two zeros at -1, but each wavelet low-pass factor has only one. Consequently, Haar has one vanishing moment, not two. In the orthogonal construction, the product's multiplicity is twice the wavelet's moment order.

The alternative names are db1 and D2: the first counts vanishing moments, while the second counts the two filter taps. We use this distinction again for the longer Daubechies filters.

13.13 The four-tap Daubechies factorization

The product filter from the handwritten example is

$$P_0(z) = \frac{1}{16}\big(-1 + 9z^{-2} + 16z^{-3} + 9z^{-4} - z^{-6}\big). \tag{13.17}$$

Only the central coefficient has an odd index, and it equals one. Therefore $P_0(z) - P_0(-z) = 2z^{-3}$. Factoring gives

$$P_0(z) = \frac{1}{16}(1 + z^{-1})^4(-1 + 4z^{-1} - z^{-2}). \tag{13.18}$$

For example, multiplying the coefficient lists $(1, 4, 6, 4, 1)$ and $(-1, 4, -1)$ produces $(-1, 0, 9, 16, 9, 0, -1)$, verifying the factorization.

Besides four zeros at -1, the product has the two roots of

$$-z^2 + 4z - 1 = 0, \qquad \alpha = 2 - \sqrt{3}, \quad \beta = 2 + \sqrt{3}, \quad \alpha\beta = 1.$$

Assign two zeros at -1 and the inside root α to G_0. Its causal reversed partner H_0 receives the outside root β. Normalizing both DC gains to $\sqrt{2}$ gives

$$G_0(z) = \frac{(1 + z^{-1})^2}{4\sqrt{2}}\big[(1 + \sqrt{3}) + (1 - \sqrt{3})z^{-1}\big], \tag{13.19}$$
$$H_0(z) = \frac{(1 + z^{-1})^2}{4\sqrt{2}}\big[(1 - \sqrt{3}) + (1 + \sqrt{3})z^{-1}\big].$$

Their residual linear factors multiply to $2(-1 + 4z^{-1} - z^{-2})$; their common denominator is 32. Their product is therefore exactly (13.18).

Expanding G_0 recovers

$$\boxed{g_0 = \frac{1}{4\sqrt{2}}(1 + \sqrt{3},\ 3 + \sqrt{3},\ 3 - \sqrt{3},\ 1 - \sqrt{3}).} \tag{13.20}$$

Here h_0 is the reversed list. Each low-pass factor has exactly two zeros at -1, so the associated orthogonal wavelet has exactly two vanishing moments. This is db2, also called D4.

The db2 zero allocation. The same product permits exchanging the two residual roots between the factors. That choice reverses the filters; it preserves magnitudes and moment orders but changes phase.

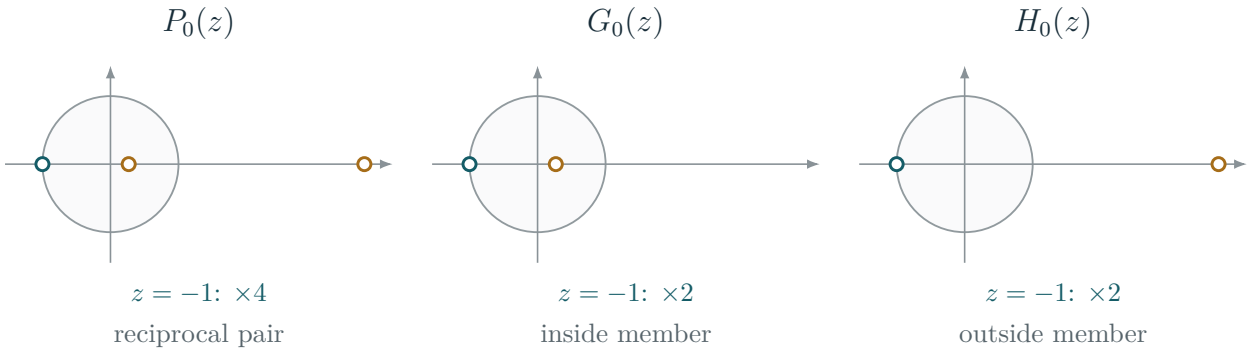


Figure 13.7: The four zeros at -1 are split equally. The remaining reciprocal pair is split between the two filters: $\alpha \approx 0.268$ and $\beta \approx 3.732$. All three panels use the same scale.

Why a linear-phase product need not have linear-phase factors. A real symmetric FIR sequence $a[0], \ldots, a[N]$ satisfies $a[n] = a[N-n]$. Its response can be written

$$A(\mathrm{e}^{\mathrm{j}\omega}) = \mathrm{e}^{-\mathrm{j}N\omega/2} \sum_{n=0}^{N} a[n] \mathrm{e}^{-\mathrm{j}\omega(n-N/2)}.$$

Pairing terms at n and $N-n$ turns the sum into real cosines. The response therefore has linear phase apart from π changes where this real amplitude changes sign.

The symmetric seven-tap product in (13.17) has Type-I linear phase and delay three. The four-tap factors are neither symmetric nor antisymmetric, so their phases are nonlinear. Haar's symmetric two-tap factors are Type II.

Minimum- and maximum-phase terminology. After removing the required factor $(1 + z^{-1})^p$, the remaining zeros of our G_0 are inside the unit circle and those of H_0 are outside. These are the customary minimum-/maximum-phase *spectral-factor choices*. The full filters have zeros on the unit circle at -1, so they do not have stable causal inverses. "Minimum phase" here must not conceal those boundary zeros.

13.14 Four moments: the product behind the db4 diagram

The eight-tap Daubechies example has $p = 4$ and $K = 7$. A concrete product underlying its zero diagram is

$$P_0(z) = z^{-7} + \frac{1}{2048}\big[-5(1+z^{-14}) + 49(z^{-2}+z^{-12}) \\ - 245(z^{-4}+z^{-10}) + 1225(z^{-6}+z^{-8})\big]. \tag{13.21}$$

Every term except z^{-7} has even exponent, so $P_0(z) - P_0(-z) = 2z^{-7}$ immediately. To read its zeros, write $v = z^{-1}$ and define

$$Q_4(v) = -5 + 40v - 131v^2 + 208v^3 - 131v^4 + 40v^5 - 5v^6.$$

Direct multiplication gives

$$\boxed{P_0(z) = \frac{1}{2048}(1+z^{-1})^8 Q_4(z^{-1}).} \tag{13.22}$$

This is an explicit algebraic version of the root diagram: eight zeros are fixed at -1, and six remain to be allocated.

Why the remaining roots occur in reciprocal-conjugate groups. The polynomial Q_4 has real, symmetric coefficients, so

$$Q_4(v)^* = Q_4(v^*), \qquad Q_4(v) = v^6 Q_4(v^{-1}).$$

Thus a nonzero root brings its conjugate and reciprocal as roots. Numerically, the three inside roots in the z-plane are

$$r_1 \approx 0.328875918, \qquad r_\pm \approx 0.284096298 \ \pm \ 0.243228226\mathrm{j}.$$

The outside roots are their reciprocals (the conjugate pair is unchanged as a set by reciprocal conjugation). All are away from the unit circle.

Choose the inside roots and four copies of -1 for G_0, and choose the outside roots and four copies of -1 for H_0. Each filter then has degree seven in z^{-1}, hence eight taps. Choosing conjugate roots together ensures real coefficients.

The db4 spectral factorization. If $\mathcal{R} = \{r_1, r_+, r_-\}$ is the inside root set, then

$$G_0(z) = c(1 + z^{-1})^4 \prod_{r \in \mathcal{R}} (1 - rz^{-1}), \qquad c = \frac{\sqrt{2}}{16 \prod_{r \in \mathcal{R}} (1 - r)}.$$

This expression fixes the gain by $G_0(1) = \sqrt{2}$. Define $H_0(z) = z^{-7} G_0(z^{-1})$. The product has exactly the zeros of (13.22); its DC gain is two, equal to $P_0(1)$. Therefore it equals P_0 identically.

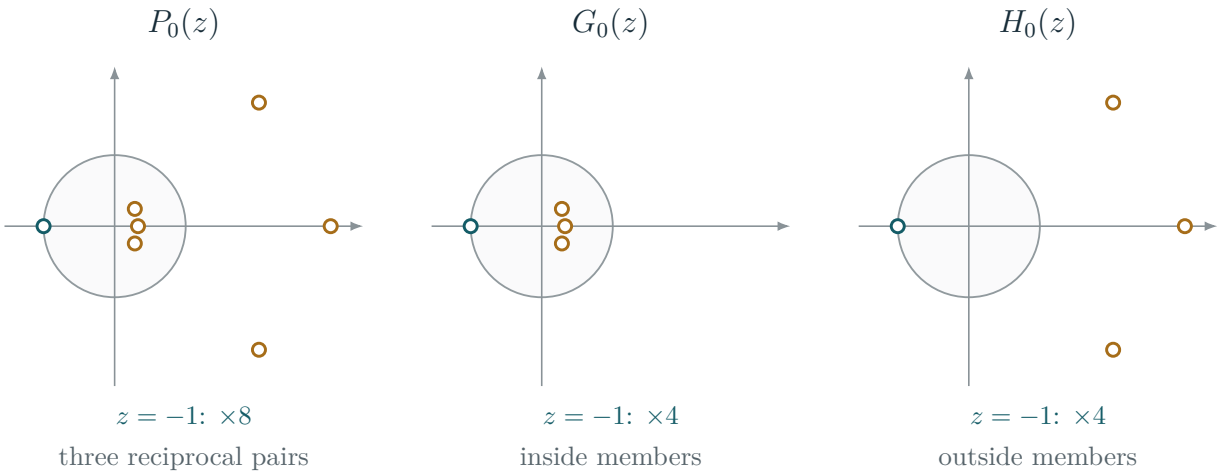


Figure 13.8: For db4 (D8), the product has eight zeros at -1. Each low-pass factor receives four. Splitting every residual reciprocal pair between the factors gives an orthogonal spectral factorization.

The product identity and the reversal relation imply

$$|G_0(\mathrm{e}^{\mathrm{j}\omega})|^2 + |G_0(\mathrm{e}^{\mathrm{j}(\omega+\pi)})|^2 = 2.$$

Together with $G_0(1) = \sqrt{2}$, this gives the orthogonal low-pass conditions. Its only unit-circle zero is -1, so the nonvanishing condition on $[-\pi/2, \pi/2]$ from Lecture 11 also holds. Thus this filter gives the orthonormal scaling function and wavelet described there. The order-four zero at -1 gives four vanishing moments.

The product is Type-I linear phase, but its eight-tap factors are not linear phase. The gain, moment order, and magnitude response do not uniquely fix phase: the placement of the remaining reciprocal roots is a genuine design choice.

13.15 Keep reciprocal pairs together to obtain symmetry

There is another allocation of the *same* product in (13.22). Give four zeros at -1 to each factor, but now place the full complex reciprocal-conjugate quartet into G_0, and the real reciprocal pair into H_0.

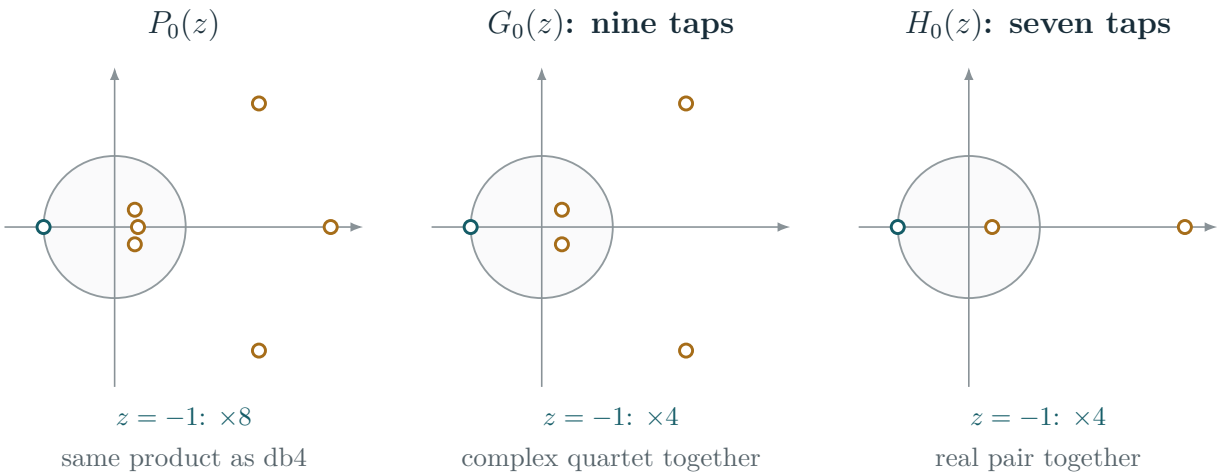


Figure 13.9: The biorthogonal 4.4 allocation keeps reciprocal pairs within a factor instead of splitting them between factors. The low-pass factors now have nine and seven taps, and each is symmetric. The product is unchanged.

Count the taps. The nine-tap factor has $4 + 4 = 8$ zeros, hence degree eight. The seven-tap factor has $4 + 2 = 6$ zeros, hence degree six. Both have four zeros at -1.

Check symmetry algebraically. A real reciprocal pair contributes

$$(1 - rz^{-1})(1 - r^{-1}z^{-1}) = 1 - (r + r^{-1})z^{-1} + z^{-2},$$

which has symmetric coefficients. A complex quartet gives the product of two such reciprocal-pair factors; including the conjugate pair makes the coefficients real. The factor $(1 + z^{-1})^4$ is also symmetric. Products of symmetric coefficient lists remain symmetric, because reversing a convolution reverses both factors. Thus both resulting filters are Type-I linear phase.

Normalize their DC gains to $\sqrt{2}$. Their product has exactly the same roots, degree, and DC gain as P_0, so it equals P_0. Perfect reconstruction still follows from (13.15), now with delay seven. The filters are no longer a reversed orthogonal pair.

A biorthogonal preview: the CDF 9/7 pair. This root allocation produces the standard biorthogonal 4.4 / CDF 9/7 low-pass pair, up to exchanging the two factors, delay choices, and normalization. Our figure labels follow the handwritten allocation: G_0 has nine taps and H_0 has seven.

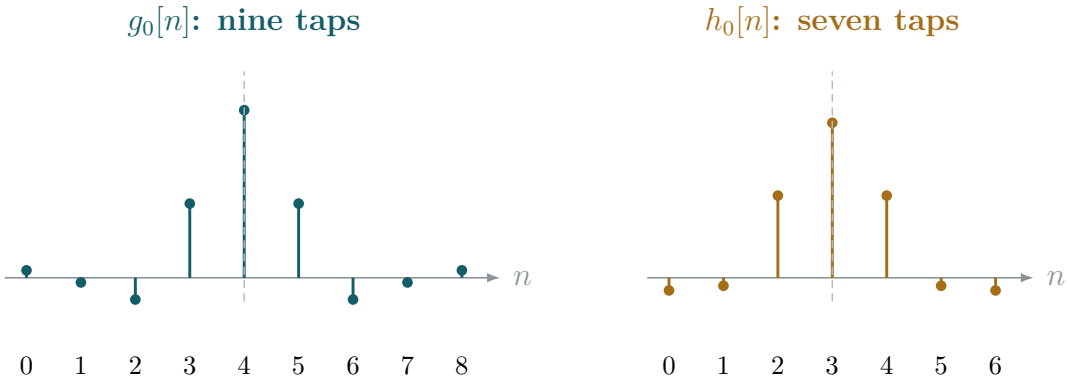


Figure 13.10: Numerical taps from the root allocation on the preceding page, with both sums normalized to $\sqrt{2}$. The centers of symmetry are four and three. Internal zeros have not been inserted to make the lists artificially equal in length.

The nine-tap analysis low-pass and seven-tap synthesis low-pass are used in the usual orientation of JPEG 2000's irreversible CDF 9/7 transform.[1] The notation 4.4 refers to four vanishing moments on each of the two wavelet sides, whereas 9/7 records the two filter lengths. This is not the same wavelet family as db4, despite the shared product polynomial.

Biorthogonal bases use two families satisfying $\langle q_\lambda, \tilde{q}_\mu \rangle = \delta_{\lambda\mu}$. In an expansion $f = \sum_\lambda c_\lambda q_\lambda$, taking the inner product with $\tilde{q}_\mu$ recovers c_μ. Thus moments of the *analyzing* wavelet control coefficient cancellation. The earlier decay proof does not use orthogonality; a non-unit norm is absorbed into $C_{\psi,p}$. Perfect reconstruction alone does not imply preservation of coefficient energy. Here we have verified the filter-bank identities; the general construction of dual MRAs is not needed for those calculations.

Pair	Low-pass lengths	Zeros at -1	Low-pass phase
Haar / db1 / D2	$2, 2$	$1, 1$	Type II
db2 / D4	$4, 4$	$2, 2$	nonlinear
db4 / D8	$8, 8$	$4, 4$	nonlinear
Biorthogonal 4.4	$9, 7$	$4, 4$	Type I

[1] This application is specified in ITU-T T.800 (2002), Annex F and Tables J.1–J.2. The standard uses a different gain normalization. Interchanging the two low-pass factors gives the dual orientation shown here.